1 GENERAL INSTRUCTIONS 1

1. Students are not allowed to leave the exam room during the first 30 minutes of the exam.
2. Late students are not permitted to complete their exams after the first 30 minutes unless authorized by the proctor (no students left the exam session), knowing that no additional time will be granted.
3. Students must use a blue or black pen. Pencils are not allowed.
4. Consumption of any food or beverage is prohibited except in medical cases.
5. All electronic devices (e.g., cell phones) are prohibited.
6. All cell phones must be turned off and placed on the invigilator's desk in the exam room.
7. All notes and reading materials must be placed on the invigilator's desk in the exam room.
8. Students are not permitted to speak to or ask questions of their classmates. If a student has a question, he or she must raise his or her hand to be assisted by the proctor.
9. Write the full information required on this page before commencing your exam.
10. Students are not allowed to leave the room once the exam commences for any reason unless it is absolutely necessary.
11. Reported student cheating cases, misbehavior, or other student misconduct on examination sessions will be reported to the Investigations Committee as per internal bylaws.

I have understood and abide with the exam policy. **Student's signature:** _____

Question No.	Question Type	CILOs assessed	Marks Weight	Scored Marks	Marker signature	Student Score
Part 1: 8	Multiple Choice	CILO 1	2 (6.67%)			
Part 1: 1-4	Multiple Choice	CILO3	8(26.67%)			
Part1: 5-7	Multiple Choice	CILO 4	6 (20%)			
Part 2: 1	Essay	CILO 3	4 (13.33%)			
Part 2: 2	Essay	CILO 4	4 (13.33%)			
Part 2: 3	Essay	CILO 2	6(20%)			
Total			30 (100%)			

Moderation Review			
Comments			
Marks Reviewed by		Signature & Date	

Part 1

Question	1	2	3	4	5	6	7	8
	C	B	B D	C	B	C	D	A

12

Name: _____ ID: A

Part 1: Choose the correct answer (16 Points)

1. The range of the function $y = \operatorname{sech} x$ is
   A. $[0,1)$ B. $(0,1]$ C. $(-1,1)$ D. $(-1,1]$

2. For any $x \in \mathbb{R}$, $\cosh(2x) =$
   A. $\cosh^2 x - \sinh^2 x$ B. $\cosh^2 x + \sinh^2 x$ C. $\sinh(2x)+1$ D. $e^{2x} + e^{-2x}$

3. $\int_0^{\ln 2} (\cosh x + \sinh x)^2 \, dx =$
   A. $\frac{1}{4}$ B. 4 C. $\frac{2}{3}$ D. $\frac{3}{2}$

4. If $f(x) = \sinh 4x$ then $f'(0) =$
   A. -4 B. 0 C. 4 D. -1

5. Making a suitable trigonometric substitution, that implies, the integral $\int \sqrt{4 - x^2}\, dx$ is equivalent to:
   A. $\int 4\cos^2\theta \, d\theta$
   B. $\int 4\cos\theta \, d\theta$
   C. $\int 2\cos\theta \, d\theta$
   D. $\int 2\cos^2\theta \, d\theta$

6. $\int x^2 \ln x \, dx =$
   A. $\frac{1}{3}x^3 \ln x + \frac{1}{9}x^3 + c$
   B. $\frac{1}{3}x^3 \ln x + \frac{1}{3}x^3 + c$
   C. $\frac{1}{3}x^3 \ln x - \frac{1}{9}x^3 + c$
   D. $\frac{1}{3}x^3 \ln x - \frac{1}{3}x^3 + c$

7. $\int_0^{\pi} x \sin x \, dx =$
   A. $\frac{\pi}{2}$ B. 0 C. $-\pi$ D. π

8. The arc length of the curve $y = \frac{2}{3}x^{\frac{3}{2}} + 1$ from $x = 0$ to $x = 1$ is equal to:
   A. $\frac{2}{3}\left(2\sqrt{2} - 1\right)$
   B. $\frac{3}{2}\left(2\sqrt{2} - 1\right)$
   C. $\frac{1}{3}\left(2\sqrt{2} - 1\right)$
   D. $\frac{1}{2}\left(2\sqrt{2} - 1\right)$

Part Two — Calculus 2 — Essay Questions — (14 Points)

1. Find : $\lim_{x\to 0^+}\left(\frac{2}{e^x-1}-\frac{2}{x}\right)$ (4 points)

$\lim_{x\to 0^+}\left(\frac{2x}{xe^x-x}-\frac{2e^x-2}{xe^x-x}\right)$ ← توحيد مقام

$\lim_{x\to 0}\left(\frac{2x-2e^x+2}{xe^x-x}\right)$ $\frac{0}{0}$

$\lim_{x\to 0}\left(\frac{2-2e^x}{xe^x+e^x-1}\right)$

$\lim_{x\to 0}\left(\frac{2-2e^0}{0+e^0-1}\right)=\frac{0}{0}$

$\lim_{x\to 0^+}\left(\frac{-2e^x}{xe^x+e^x+e^x}\right)\to\left(\frac{-2e^x}{xe^x+2e^x}\right)\to\frac{-2}{2}=\boxed{-1}$ 4

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2. Evaluate: $\int \sin^3 x\cos^2 x\,dx$ (4 points)

if at least one is odd — case 1

$\sin^2x+\cos^2x=1$

$\int \sin x\cdot\sin^2x\cdot\cos^2x\,dx$

$\int \sin x\cdot(1-\cos^2x)\cdot\cos^2x\,dx$

$u=\cos x$

$\frac{du}{-\sin x}=dx$

$\int \sin x\cdot(1-u^2)\cdot u^2\frac{du}{-\sin x}$

$-\int(1-u^2)(u^2)\,du$

$-\int u^2-u^4\,du$

$-\left[\frac{u^3}{3}-\frac{u^5}{5}\right]+C$ 4

$\frac{u^5}{5}\quad\frac{-u^3}{3}$

$u=\cos x$

$\frac{\cos^5x}{5}-\frac{\cos^3x}{3}+C$

$$V = \pi \int_a^b f(x)^2 \, dx$$

3. Find the volume of the solid that is obtained when the region that is in the first quadrant and bounded by $y = \sqrt{25 - x^2}$ and $y = 3$, is rotated a complete revolution about the x-axis. (6 Points)

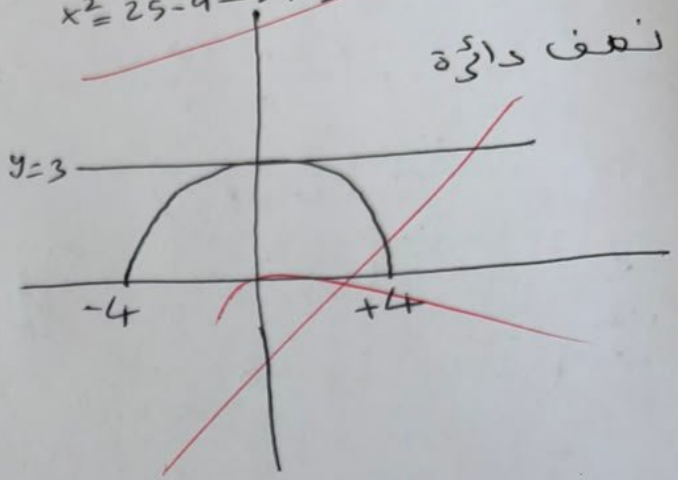
(Sketch the region that will be rotated)

$3 = \sqrt{25 - x^2}$

$9 = 25 - x^2$

$x^2 = 25 - 9 \rightarrow x = \pm 4$

نصف دائرة



$$\pi \int_{-4}^{4} \left(\sqrt{25 - x^2}\right)^2 dx$$

$$\pi \int_{-4}^{4} 25 - x^2 \, dx$$

$$\pi \left[25x - \frac{x^3}{3} \Big|_{-4}^{4} \right]$$

$$\pi \left[\left(100 - \frac{64}{3}\right) - \left(-100 + \frac{64}{3}\right) \right]$$

$$\pi \left[100 - \frac{64}{3} + 100 - \frac{64}{3} \right]$$

$$\pi \left[200 - \frac{128}{3} \right]$$

$$V = \frac{\pi \, 442}{3}$$

Good Luck